

InstaSHAP One-Page Summary

Objective: reproduce InstaSHAP on Bike Sharing, Covertypes, and Adult Income.

Core method: train an additive model with the paper's masked objective so SHAP-like attributions can be read directly from additive components in one forward pass.

Available result tables:

- adult: best blackbox accuracy=0.8400
- bike: best gam2 nmse_pct=187.4328
- covertime: best gam2 accuracy=0.6913

Artifacts generated by the project:

- Dataset loaders and preprocessing using ucimlrepo
- Black-box, GAM, surrogate, and InstaSHAP implementations
- Metrics tables, plots, PDF report, and notebook scaffold